

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY COURSE
CODE: CSA5115

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

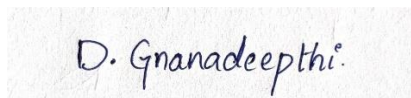
Total Marks: 50

Annexure A

Code of Conduct:

I D. Gnana Deepthi(192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.

B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.
- B. Recommend a suitable cryptographic approach and justify your answer.

ANSWERS

Scenario-Based 3: Key Exchange in an Unsecured Network

A. Explain how the Diffie-Hellman key exchange works

Diffie-Hellman Key Exchange is a cryptographic method that allows two users to establish a **shared secret key** over an unsecured public network without transmitting the secret key directly.

Steps:

1. Both users publicly agree on a **prime number (p)** and a **generator (g)**.
2. **User A** selects a private key **a** and computes the public key:

$$A = g^a \text{ mod } p$$

3. **User B** selects a private key **b** and computes the public key:

$$B = g^b \text{ mod } p$$

4. The users exchange their public keys over the public network.
5. **User A** computes the shared secret key:

$$K = B^a \text{ mod } p$$

6. **User B** computes the shared secret key:

$$K = A^b \text{ mod } p$$

Both users obtain the **same shared secret key**, which is then used to encrypt and decrypt their communication securely.

B. Identify a possible security threat and suggest a solution

Security Threat: Man-in-the-Middle (MITM) Attack

A major threat to Diffie-Hellman is the **Man-in-the-Middle (MITM) attack**. In this attack, an attacker intercepts the public keys exchanged between the two users and replaces them with their own public keys. As a result, each user unknowingly establishes a separate secret key with the attacker instead of with each other. The attacker can then read, modify, and forward all messages without either user realizing it.

Solution to Mitigate the Attack

The attack can be prevented by **authenticating the public keys** before they are used. This can be achieved by:

- Using **Digital Certificates** issued by a trusted **Certificate Authority (CA)**.
- Using **Digital Signatures** to verify the identity of the communicating users.
- Implementing **Public Key Infrastructure (PKI)** or authenticated Diffie-Hellman protocols (such as those used in TLS).

These methods ensure that the exchanged public keys are genuine and prevent attackers from impersonating legitimate users.

Scenario-Based 4: Public Key Cryptography for Email Security

A. Describe how RSA can be used for Encryption and Decryption

RSA is a **public key cryptography** algorithm that uses two keys:

- **Public Key** – used for encryption.
- **Private Key** – used for decryption.

Working of RSA in Email Security

1. Each employee generates an RSA **public key** and **private key**.
2. The sender obtains the recipient's **public key**.
3. The sender encrypts the email using the recipient's public key.
4. The encrypted email is sent over the network.
5. The recipient decrypts the email using their **private key**.
6. Since only the recipient possesses the private key, only they can read the email.

Result: The email remains confidential even if it is intercepted during transmission.

B. Explain how Key Distribution and Management Maintain Security

Key distribution and management are essential for the secure use of RSA.

Key Distribution

- Public keys should be shared through a **trusted source**, such as a **Public Key Infrastructure (PKI)** or **Digital Certificates**.
- This ensures users receive the correct public key and prevents attackers from substituting fake keys.

Key Management

- Private keys must be kept **secret** and stored securely.
- Public keys should be updated and verified regularly.
- Lost or compromised keys should be **revoked** and replaced immediately.
- Organizations should use **certificate authorities (CAs)** to issue and manage digital certificates.

Importance:

- Prevents unauthorized access.
- Protects against Man-in-the-Middle (MITM) attacks.
- Ensures confidentiality, authenticity, and secure communication.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well reasoned, logical decisions	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided

		with strong rationale			
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand